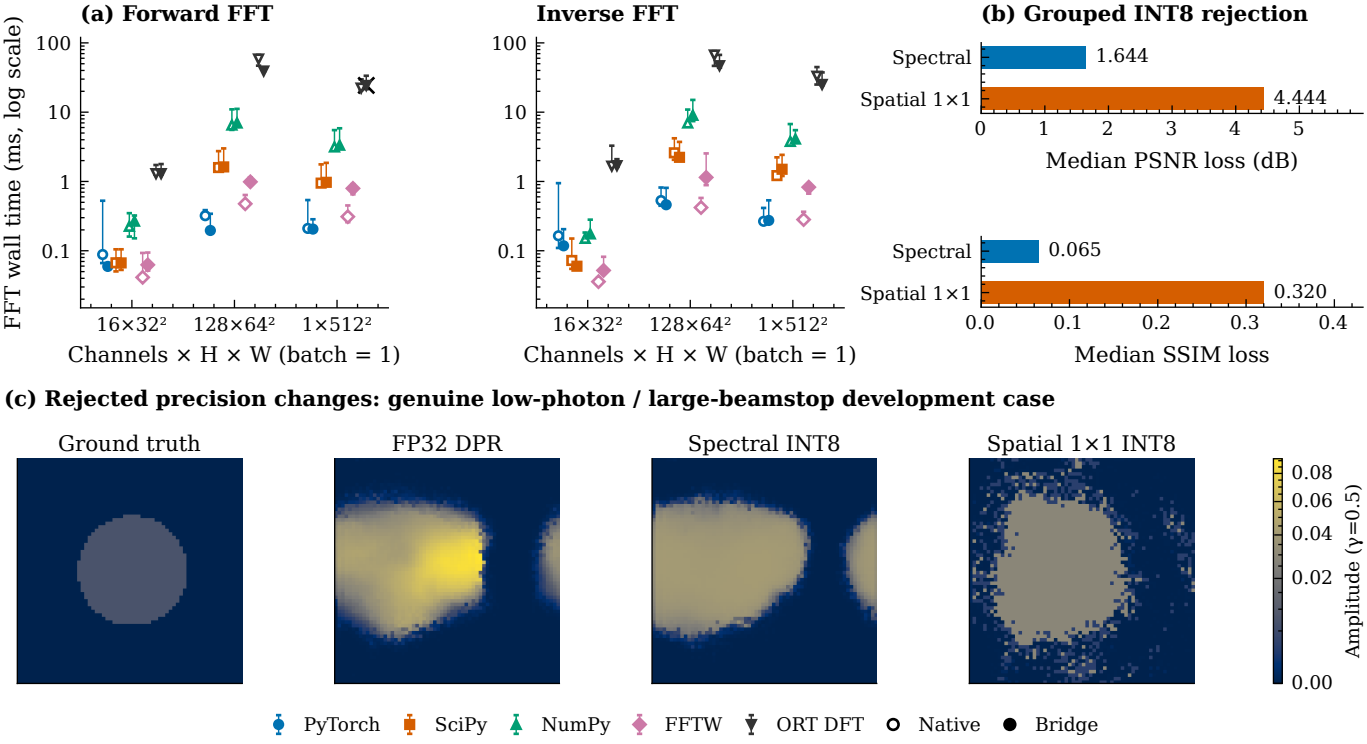

CPU kernels and reconstruction fidelity: completed technical comparisons
(a) Forward FFT
Inverse FFT
(b) Grouped INT8 rejection
FFT wall time (ms, log scale)
100
10
1
0.1
16×32²
128×64²
1×512²
Channels × H × W (batch = 1)
Spectral
Spatial 1×1
1.644
4.444
0
1
2
3
4
5
Median PSNR loss (dB)
0.065
0.320
0.0
0.1
0.2
0.3
0.4
Median SSIM loss
(c) Rejected precision changes: genuine low-photon / large-beamstop development case
Ground truth
FP32 DPR
Spectral INT8
Spatial 1×1 INT8
0.08
0.06
0.04
0.02
0.00
Amplitude (γ=0.5)
PyTorch
SciPy
NumPy
FFTW
ORT DFT
Native
Bridge